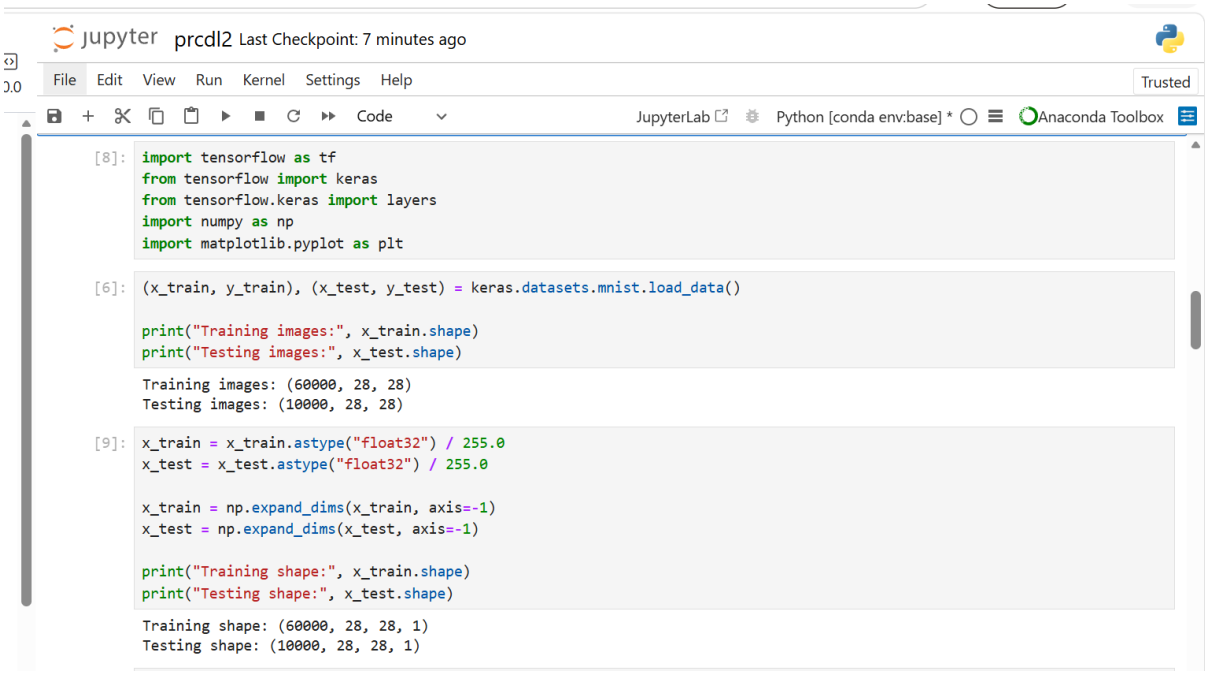
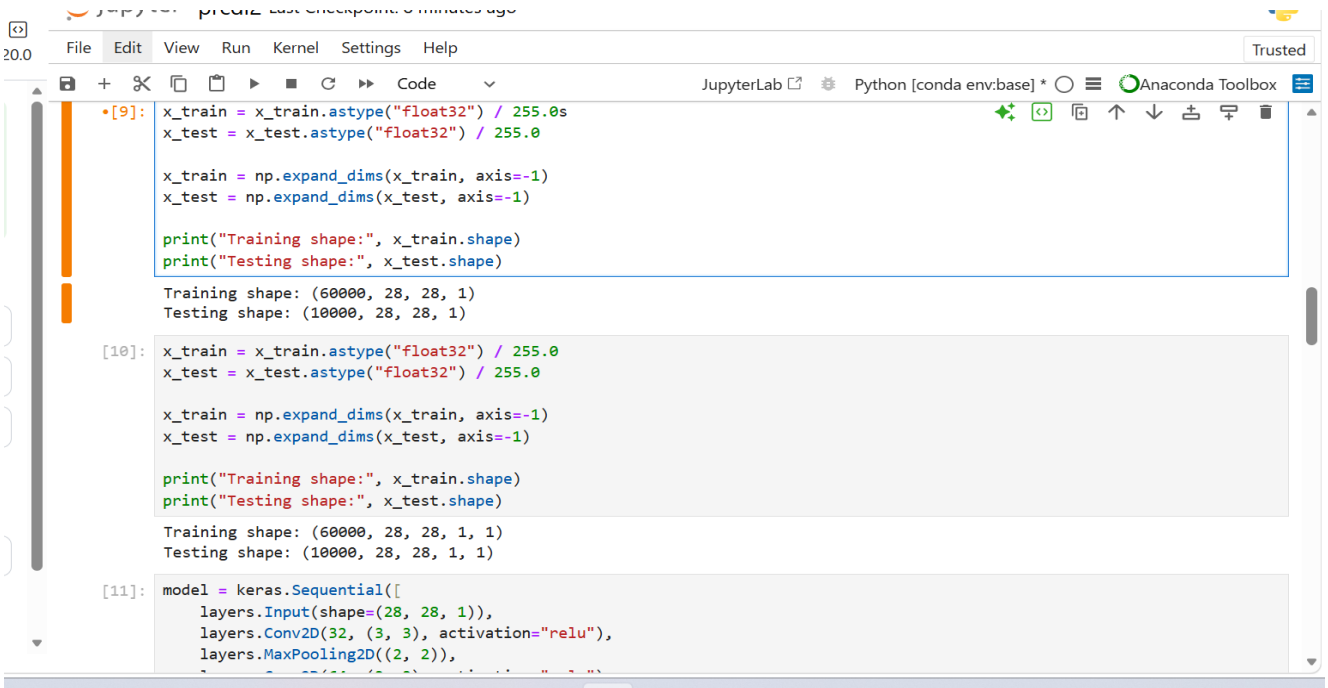

jupyter prcdl2 Last Checkpoint: 6 minutes ago
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Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
[7]: %pip install tensorflow numpy matplotlib
Requirement already satisfied: tensorflow in c:\users\anjali\anaconda3\lib\site-packages (2.21.0)
Requirement already satisfied: numpy in c:\users\anjali\anaconda3\lib\site-packages (1.26.4)
Requirement already satisfied: matplotlib in c:\users\anjali\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (2.5.1)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (24.1)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (2.32.3)
Requirement already satisfied: setuptools in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (75.1.0)
Requirement already satisfied: six>=1.12.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.14.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.84.1)


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Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
[8]: import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import numpy as np
import matplotlib.pyplot as plt
[6]: (x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
print("Training images:", x_train.shape)
print("Testing images:", x_test.shape)
Training images: (60000, 28, 28)
Testing images: (10000, 28, 28)
[9]: x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)
print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)
Training shape: (60000, 28, 28, 1)
Testing shape: (10000, 28, 28, 1)


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Code
JupyterLab
Python [conda env:base] *
Anaconda Toolbox
•[9]: x_train = x_train.astype("float32") / 255.0s
x_test = x_test.astype("float32") / 255.0
x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)
print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)
Training shape: (60000, 28, 28, 1)
Testing shape: (10000, 28, 28, 1)
[10]: x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)
print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)
Training shape: (60000, 28, 28, 1, 1)
Testing shape: (10000, 28, 28, 1, 1)
[11]: model = keras.Sequential([
layers.Input(shape=(28, 28, 1)),
layers.Conv2D(32, (3, 3), activation="relu"),
layers.MaxPooling2D((2, 2)),

```
    layers.Flatten(),
    layers.Dense(128, activation="relu"),
    layers.Dense(10, activation="softmax")
])

model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 128)	204,928
dense_1 (Dense)	(None, 10)	1,290

Total params: 225,034 (879.04 KB)
Trainable params: 225,034 (879.04 KB)
Non-trainable params: 0 (0.00 B)

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
import matplotlib.pyplot as plt
```

[6]:
```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

print("Training images:", x_train.shape)
print("Testing images:", x_test.shape)
```

```
Training images: (60000, 28, 28)
Testing images: (10000, 28, 28)
```

[9]:
```
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)

print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)
```

```
Training shape: (60000, 28, 28, 1)
Testing shape: (10000, 28, 28, 1)
```

[10]:
```
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

x_train = np.expand_dims(x_train, axis=-1)
x_test = np.expand_dims(x_test, axis=-1)
```

```
844/844 ━━━━━━━━━━━━━━━━━━━━ 15s 17ms/step - accuracy: 0.9683 - loss: 0.1031 - val_accuracy: 0.9755 - val_loss: 0.0894
```

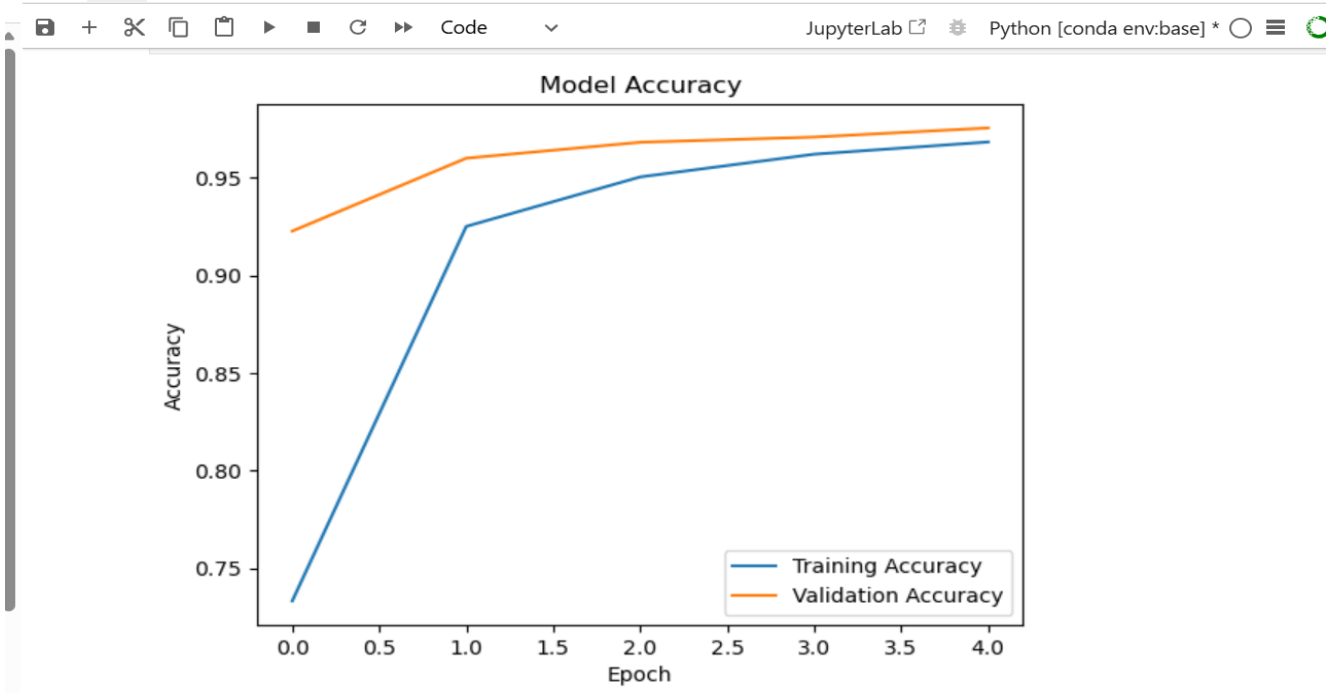
[14]:
```
test_loss, test_accuracy = model.evaluate(x_test, y_test)

print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy * 100, "%")
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.9729 - loss: 0.0905
Test Loss: 0.09048764407634735
Test Accuracy: 97.28999733924866 %
```

[15]:
```
plt.plot(history.history["accuracy"], label="Training Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")

plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Model Accuracy")
plt.legend()
plt.show()
```



Epoch

[16]:
```
plt.plot(history.history["loss"], label="Training Loss")
plt.plot(history.history["val_loss"], label="Validation Loss")

plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Model Loss")
plt.legend()
plt.show()
```

